

CSE 5830: Probabilistic Graph Analysis

Homework 3

Due 11:59PM, Wednesday 9/30/2026

Show all your work to get a full score.

Problem 1: Noisy-AND (15 points)

A Noisy-AND model (independence of the causal mechanisms when all X s are 1) with binary random variables and leaky parameters $(\lambda_0, \lambda_1, \dots, \lambda_K)$.

(a) Expression for $P(Y = 0 \mid X_1, \dots, X_K)$ (10 points)

Setup / convention. [TODO: State what λ_i and λ_0 mean in your model (in words), and draw or describe the auxiliary variables Z_i if you use them.]

Derivation. [TODO: Derive the expression step by step.]

$$P(Y = 0 \mid X_1, \dots, X_K) = [\text{TODO: final expression}]$$

(b) Numerical case (5 points)

$$K = 4, \quad \lambda_0 = 0, \quad \lambda_1 = 0.9, \quad \lambda_2 = 0.75, \quad \lambda_3 = 0.8, \quad \lambda_4 = 0.85.$$

$$P(Y = 0 \mid X_1 = 1, X_2 = 0, X_3 = 0, X_4 = 1) = [\text{TODO: substitute into (a)}] = \underline{\hspace{2cm}}$$

[TODO: One sentence: does the number make sense? (sanity check)]

Problem 2: Factor product (15 points)

Given factors:

X	Y	$\phi_1(X, Y)$	Y	Z	$\phi_2(Y, Z)$
1	1	0.8	1	1	0.2
1	2	0.5	1	2	0.2
2	1	0.5	2	1	0.9
2	2	0.6	2	2	1.0

(a) $\psi(X, Y, Z) = \phi_1(X, Y) \phi_2(Y, Z)$ (10 points)

X	Y	Z	$\phi_1(X, Y) \cdot \phi_2(Y, Z)$	$\psi(X, Y, Z)$
1	1	1	_____ . _____	_____
1	1	2	_____ . _____	_____
1	2	1	_____ . _____	_____
1	2	2	_____ . _____	_____
2	1	1	_____ . _____	_____
2	1	2	_____ . _____	_____
2	2	1	_____ . _____	_____
2	2	2	_____ . _____	_____

(b) $\psi(X, Z \mid Y = 1)$ (5 points)

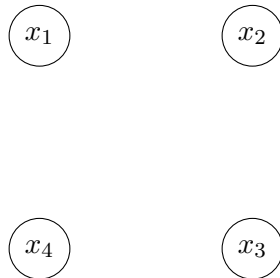
[TODO: Explain in one or two sentences what operation you are applying (factor reduction to the context $Y = 1$) and whether you normalize.]

X	Z	$\psi(X, Z \mid Y = 1)$
1	1	_____
1	2	_____
2	1	_____
2	2	_____

Problem 3: Pairwise Markov network (10 points)

$$p(\mathbf{x}) \propto \phi(x_1, x_2) \phi(x_2, x_3) \phi(x_3, x_4) \phi(x_4, x_1)$$

(a) Draw the network (2 points)



(b) Conditionals in terms of ϕ (8 points)

[TODO: Show the derivation once in full for $p(x_1 \mid x_2, x_4)$ (write the conditional as a ratio of joints, then cancel), then use the same pattern for the rest.]

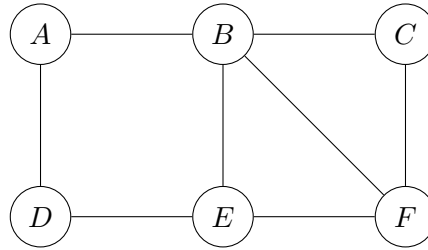
$$p(x_1 \mid x_2, x_4) = [\text{TODO: }]$$

$$p(x_2 \mid x_1, x_3) = [\text{TODO: }]$$

$$p(x_3 \mid x_2, x_4) = [\text{TODO: }]$$

$$p(x_4 \mid x_1, x_3) = [\text{TODO: }]$$

Problem 4: Markov network structure (20 points)



(a) All factors that factorize over the graph (6 points)

Edges: **[TODO: list them]**

Cliques: **[TODO: list every clique (single nodes, edges, triangles, ...)]**

Maximal cliques: **[TODO:]**

$$P(A, B, C, D, E, F) = \frac{1}{Z} [\text{TODO: product of factors}]$$

(b) Gibbs distribution as a pairwise MN (6 points)

$$P(A, B, C, D, E, F) = \frac{1}{Z} [\text{TODO: product of pairwise factors}], \quad Z = [\text{TODO: }]$$

(c) Local Markov independencies (6 points)

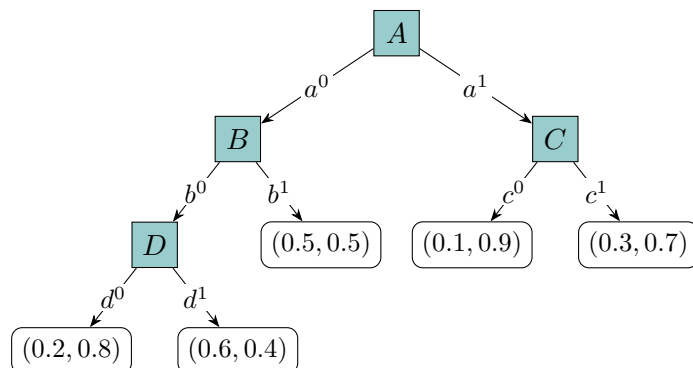
Node	Neighbors (Markov blanket)	Local independence
A	_____	[TODO:]
B	_____	[TODO:]
C	_____	[TODO:]
D	_____	[TODO:]
E	_____	[TODO:]
F	_____	[TODO:]

(d) Is $A \perp F \mid \{D, C\}$? (2 points)

[TODO: Answer yes/no, and justify with separation: either exhibit an active path from A to F that avoids {D, C}, or argue that every path passes through {D, C}.]

Problem 5: Tree-structured CPD (10 points)

Leaves list $(P(e^0), P(e^1))$.



(a) $P(E = e^0 \mid a^0, b^1, c^1, d^1)$

Path through the tree: **[TODO:]** Answer: _____

(b) $P(E = e^1 \mid a^1, b^1, c^1, d^1)$

Path through the tree: **[TODO:]** Answer: _____

(c) $P(E = e^1 \mid a^0, b^0, c^0, d^0)$

Path through the tree: **[TODO:]** Answer: _____

(d) Is $E \perp_c B \mid a^0$?

[TODO: Yes/no + justification using the tree.]

(e) Is $E \perp_c C \mid d^1$?

[TODO: Yes/no + justification using the tree.]